

CSC186 – Object Oriented Programming
Academic Session March 2025 – August 2025
Lab Assignment 3 – Concept of Classes II

Course Outcomes (CO)	LO1	LO2	LO3
CO1			
CO2	√	√	√
CO3			

Write a program using Java Programming Language

QUESTION 1

Given the following class `Cloth` consists of the following attributes:

- Name //customer's name. e.g: Siti Salwa Ahmad
- Price //price of the cloth per meter. e.g: 15.50
- Length //length of the cloth in meter: e.g: 2.5

a) Write a complete class for the following methods:

- i) A default/normal/copy constructor.
- ii) Mutator for each attribute.
- iii) Accessor for each attribute.
- iv) A printer method named `toString()` to return and display information of the object.
- v) Processor method named `calcPayment()` to calculate and return the price to be paid by the customer based on price per meter and length of the cloth. The 6% of GST will be imposed from the price paid.

b) Write an application program that performs the following:

- i) Declare an array of the object for `cloth` object.
- ii) Input some data in the array of the object.
- iii) Display all customer information, including the payment price.
- iv) Calculate and display the total payment paid by customers.
- v) Count and display the number of customers who buy the cloth more than 10 meters.
- vi) Find and display the customer details, who made the highest payment.

QUESTION 2

The appeal process of paper rechecking is opened for two weeks after the result is announced to students. The charge of paper rechecking is RM 50 per paper. The following class shows the data members of `PaperRechecking` class.

```
public class PaperRechecking {  
    //attributes :  
    private String studID; //student id  
    private String studName; //student name  
    private String subjectCode; // subject code, e.g: CSC435  
    private int previousMark; //the mark obtained before the  
                                //rechecking process  
    private int newMark; // the mark obtained after the  
                                // rechecking process  
  
    //methods :  
    Constructors  
    Mutators  
    Retrievers  
    Processor  
    Printer  
}
```

- a) Write a function definition for each of the following questions:
- Default/Normal/Copy Constructor
 - Setter/Mutator for each attribute
 - Getter/Retriever/Accessor for each attribute
 - A Processor named `DisplayChanges()` which display the changes of mark. The following table shows the appropriate message that should be displayed.
 - Printer method using `toString()` defined method.

Previous Mark	New Mark	Message
Higher	Lower	The mark after recheck deducted (previousMark - newMark)
Lower	Higher	The mark after recheck added (newMark - previousMark)
Same	Same	No changes of mark

- b) Write a program segment in the main program for each of the following questions.
- Create an array of object named `pRechecking` and the array size is 100. Then, input some data into array of objects.
 - Display all subject codes, the changes of marks, total codes and total charges, rechecked by a student named "Ali bin Hassan".
 - Count and display the number of students who obtained the same marks after the rechecking process.

QUESTION 3

Class `Land` has the following attributes and methods:

Attributes:

- `id`
- `owner name`
- `house type`
- `area`

a) Write the `Land` class and the following methods:

- Default/Normal/Copy constructor.
- Mutator/Setter method for each attribute
- Retriever method for each attribute.
- Printer method using `toString()` defined method.
- A processor method to calculate and return the tax amount. The tax of this type of land depends on its area, and the type of house built on the land as shown in the following table:

House Type	Description	Tax rate (RM/m ³)
T	Terrace	10
S	Semi-Detached	15
B	Bungalow	20
C	Condominium	30

Details of land

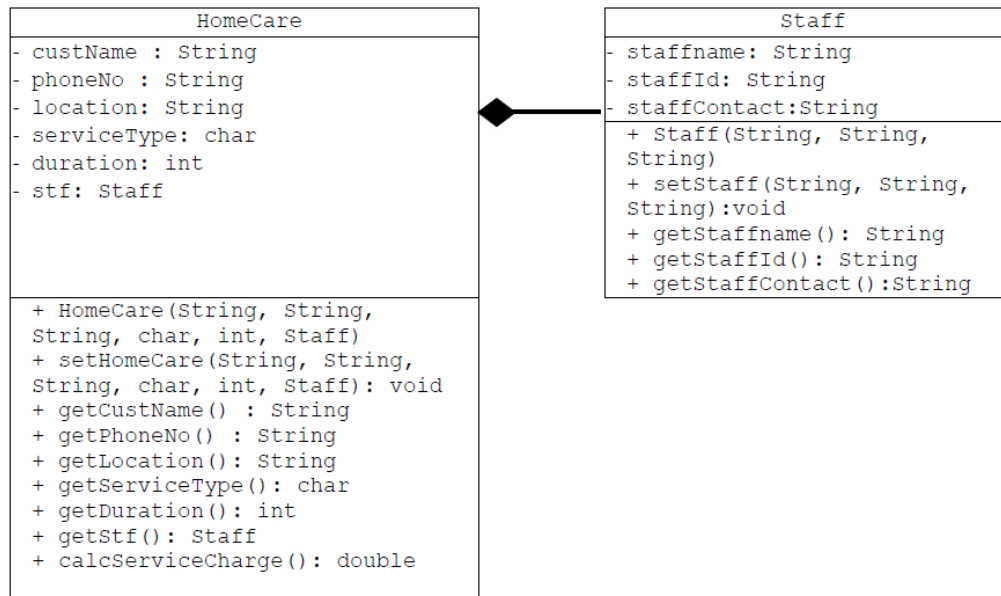
b) Write an application program that performs the following:

- Declare array of object for `Land` objects.
- Input some data onto the array of objects.
- Display the land information from the Condominium house type.
- Calculate and display the total price for each type of the house.
- Calculate and display the average tax price for Semi-Detached house type.
- vi) Determine and display the highest tax price for Bungalow house type.

QUESTION 4

NurseAtHome Sdn Bhd is a professional home health care company that provide quality home health care services including basic personal care, physiotherapy and baby care. The company is committed to offer the best services to the customer. In order to make the service more organized, especially in keeping track of customer and nurse service records, the company is planning to upgrade the existing system. Thus, you are assigned to develop an application using object-oriented approach.

The following UML class diagram stated the information needed in this development process.



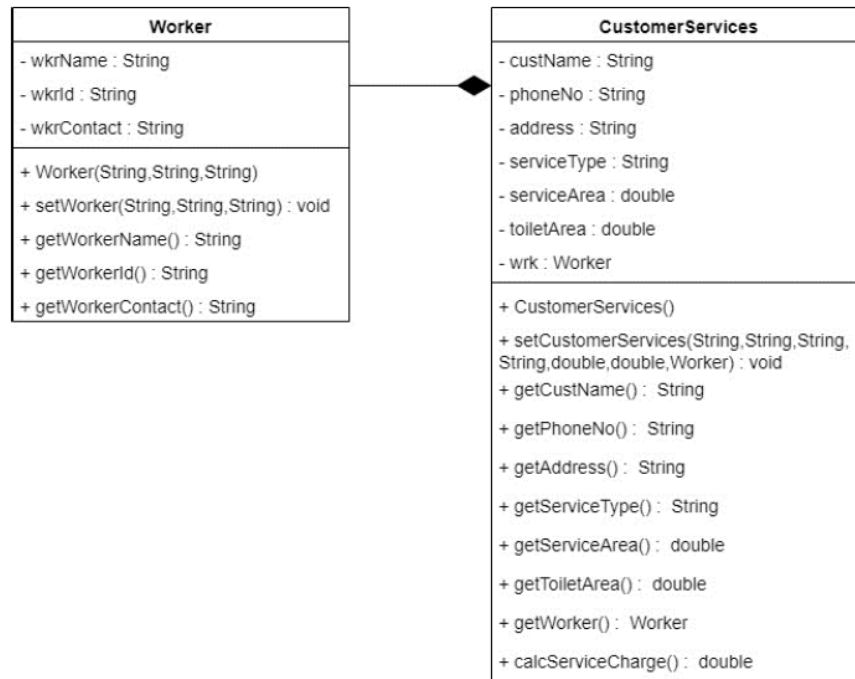
- a) Write a complete class for Homecare and Staff.
- i) Write a method named `calcServiceCharge()`. The total service charge is calculated based on the service type and service hour. A customer will receive a discount 5% if service more than 5 hours. The table below shows the rate of service charge imposed on customers.

Service Type	Service Type	Price per hour (RM)
P	Basic Personal Care	60.00
T	Physiotherapy	120.00
B	Baby Care	80.00

- b) Write Java main program (Java Application Program) segments to do the following tasks:
- i) Create an array named `assist` to store 30 `HomeCare` objects. Then, read and store all data into the array of objects.
- ii) Calculate and display total charges for customers who choose physiotherapy service.

QUESTION 5

abCLEAN is a cleaning company that provides cleaning services such as grass cutting, room cleaning, and toilet cleaning. The toilet cleaning service can only be requested if the customer chooses the room cleaning service. In order to provide efficient services, the company is planning to upgrade the existing record system. The new system will be able to store customer information, worker information, service information and calculate service costs. Thus, you are assigned to develop an application using an object-oriented approach. The figure below shows a class diagram of the new system.



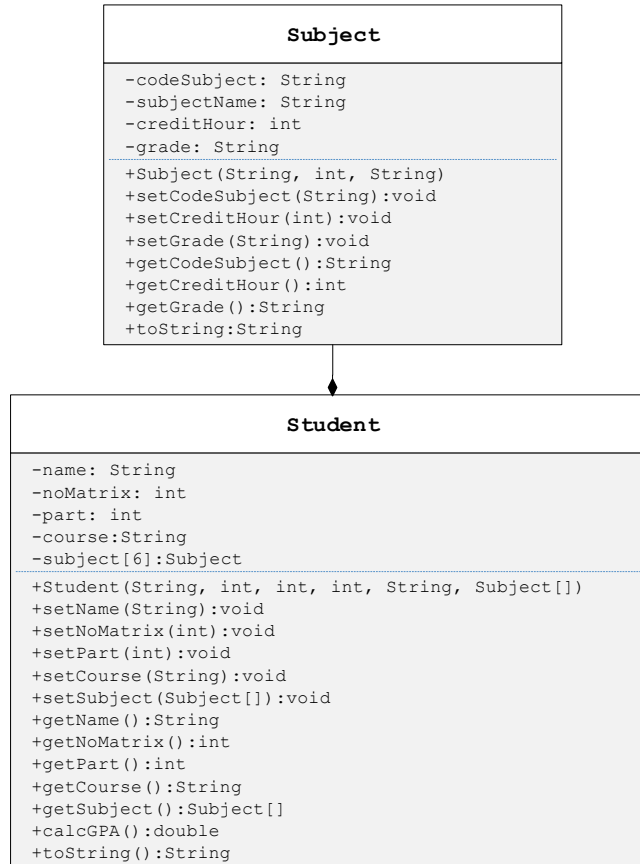
- a) Write a complete class for `Worker` and `CustomerServices`.
- i) Write a method named `calcServiceCharge()` to calculate and return the total service cost that needs to be paid by the customer. A customer will receive a discount of 3% if the service cost is more than RM150. The table below shows the rate of service costs imposed on customers.

Service Type	Price per feet ² (RM)
Grass	0.85
Room	2.00
Toilet	5.00

- b) In the Java main program (Java Application Program), answer the following questions.
- i) Create an array to store **FIFTY (50)** `CustomerServices` object. Then, read all data and store it into the array of object.
- ii) Calculate and display the total service charges collected from all customers for each type of service.
- iii) Display the IDs of workers who have done the grass-cutting job.
- iv) Calculate and display the average service charges collected from customers.

QUESTION 6

Write a program to calculate a student's Grade Point Average (GPA). Each student is enrolled in 6 subject per semester. Given the following class diagram as follows:



- a) Write a complete class for Subject and Student.
- i) Write a method named `calcGPA()` to calculate and return the Grade Point Average (GPA). The table below shows the value of the grade

Grade	Grade Point
A+	4.00
A	4.00
A-	3.67
B+	3.33
B	3.00
B-	2.67
C+	2.33
C	2.00
C-	1.67
D+	1.33
D	1.00
E	0.67
F	0.00

Steps to Calculate GPA

1. **Convert Grades to Points**
For each course, convert your letter grade to the corresponding point value.
2. **Multiply by Credit Hours**
Multiply the grade point by the number of credit hours for the course.
3. **Add Up**
 - Total all the grade points earned (grade point × credit hours).
 - Total all the credit hours attempted.
4. **Divide**
Divide the total grade points by the total credit hours.

☑ Example

Subject	Grade	Grade Point	Credit Hours	Quality Points (GP × CH)
Math	A	4.00	3	12.00
English	B+	3.33	3	9.99
History	B	3.00	4	12.00
Science	A-	3.67	2	7.34

- Total Credit Hours: 12
- Total Quality Points: 41.33

✚ **GPA = Total Quality Points / Total Credit Hours = 41.33 / 12 = 3.44**

b) Write an application program that performs the following:

- i) Declare an array of the object for `Student` object. Then, Input some data onto the array of object
- ii) Display the detailed information of each student as follows:

Student Name : Abu Bakar bin Mahmud
No. Matrix : 2023653997
Part : 2
Course : CS110 (Diploma in Computer Science)

Code	Subject	Subject	Grade	Grade Point	Credit Hours	Quality Points (GP × CH)
MAT123		Math	A	4.00	3	12.00
CCU127		English	B+	3.33	3	9.99
HST101		History	B	3.00	4	12.00
SCI201		Science	A-	3.67	2	7.34
CSC221		Programming	A	4.00	3	12.00
SCU224		Security	B-	2.67	3	8.01

Total Credit Hours: 18
GPA = 3.41

- iii) Calculate the number of students who get GPA greater than 3.00
- iv) Display the student information who gets Dean List. Dean list is given based on the GPA greater or equal to 3.67
- v) Find the highest GPA and display student information.